

Yingrui Hou

Simulation Engineer - Physical Systems, Calibration and Validation

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SUMMARY

I build calibrated models from noisy measurements and imperfect simulations for physical systems.

I build calibrated models from noisy measurements and imperfect simulations. My core strength is designing modelling, calibration, and validation strategies that extract reliable conclusions when the true state is only indirectly observed.

In postdoctoral work, that meant building reusable C++/Python/ROOT/Geant4 workflows for simulation-to-test alignment, calibration, monitoring, time-dependent inference, and model correction in large shared technical environments. The value came from turning validation results into calibration updates, software follow-up actions, and more credible modelling decisions.

CORE SKILLS

- Simulation and modelling: ROOT / RooFit, Geant4, Simulation-to-Data Alignment, Signal Processing
- Calibration and validation: Model Validation, Calibration & Bias Correction, Domain Adaptation
- Data and inference: Probabilistic Modelling, Statistical Inference, Maximum Likelihood Estimation, Large-Scale Data Analysis
- Engineering workflow: Python, C/C++, Workflow Automation (YAML / Shell / CMake)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.
- Worked in large technical collaborations with shared codebases and review discipline, with validation outputs feeding calibration and model decisions.
- Built and adapted Geant4/C++ simulation code for optical-response and detector studies, including geometry, stepping, and analysis outputs.
- Combined beam-test data, simulation, and response maps to model measurement-chain behaviour and quantify non-uniformity effects on system resolution, including a quantified constant-term contribution below 1%.
- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.
- Built benchmark-based monitoring workflows to validate reconstruction and calibration quality, then traced anomalies to detector behaviour or software issues instead of stopping at descriptive analysis.
- Implemented event-level model-correction tooling inside collaboration frameworks to improve fast-model realism using measured constraints.

SELECTED OUTCOMES

- Quantified response non-uniformity contribution to resolution below 1% in a detailed detector study
- Improved reconstructed-energy resolution by about 2x in representative PSD-based studies
- Built validation workflows that exposed detector and software issues, including a missing-energy bug later fixed downstream

TECHNICAL HIGHLIGHTS

- Model validation and uncertainty quantification for noisy measurement systems
- Simulation-to-measurement alignment through calibration, reweighting, and correction workflows
- Sensor and system modelling with Geant4, ROOT, Python, and C++
- Test-data analytics, anomaly tracing, and root-cause diagnosis
- ML validation under dataset shift for decision-facing pipelines

INDUSTRIAL FIT

Best fit for industrial R&D teams working on physical systems, especially where simulation-to-measurement alignment, calibration, model validation, and uncertainty-aware analysis are central to the work. Strongest value comes when the goal is not generic analytics, but making models trustworthy enough to support engineering decisions.

Across these projects, the recurring pattern is consistent: build the workflow, correct mismatch between simulation and measurement, validate aggressively before trusting the result, and make the final model usable for decisions rather than just analysis. That is the same pattern behind industrial R&D teams working on physical systems, test-data interpretation, monitoring, calibration, model credibility, and simulation-informed design iteration.

EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2